Reinforcement Learning vs. Rule-Based Policies for Dynamic Credit Threshold Control: A Simulation Study

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Abstract

Portfolio-level credit acceptance thresholds are revised periodically in practice, which makes the decision sequential rather than one-shot. Each week the controller sets two acceptance thresholds, one for new clients and one for repeat clients, and then observes delayed repayment and default outcomes over subsequent weeks. We evaluate six deep reinforcement learning (RL) algorithms (DQN, Double-DQN, PPO, SAC, A3C, and A2C) against five rule-based baselines in a calibrated synthetic credit-portfolio simulator across six market scenarios and four observation-space sizes (12D, 20D, 30D, 50D). Data-driven weekly threshold control is one concrete way intelligent systems can improve business lending decisions, because adjusting thresholds to current portfolio and market signals can protect profit as conditions change, and this study measures how close learned controllers come to well-

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designed rules. The simulator is a calibrated proof-of-concept test bench, not a substitute for real banking data. Only observation dimensionality varies across the four conditions, with the simulator, reward, seeds, and evaluation protocol held fixed. We find that the constraint-aware weekly policy remains the best overall controller by cumulative reward (the primary ranking metric, with realized profit 3,802,547) and is within 0.02% of the profit-oriented policy on realized profit, and RL does not win the reward or the profit ranking in any scenario or dimension. Agents are trained for 1,500 episodes per seed, a substantially larger budget than a short pilot run, and under this budget the best RL agent closes most of the profit gap to the leading baseline, from -9.4% under a short 50-episode budget to between -1.0% (12D, SAC) and -2.4% (50D, SAC), with the gap widening as the observation grows. The strongest RL agent varies by dimension (SAC at 12D, 20D, and 50D, A2C at 30D), and the margins between the leading RL agents are smaller than their seed-to-seed spread, so no single algorithm dominates. A controlled state-dimensionality comparison on the canonical DQN agent shows an intermediate optimum: cumulative reward rises from 12D to 20D and 30D (20D is significantly ahead of 30D, and 30D of 12D) and collapses at 50D. Richer state does not buy a risk-side advantage: the best RL agent's default rate stays slightly above that of the leading baseline at every dimension, and it undercuts both profit-seeking baselines on default rate in at most two of six scenarios. A delayed-evaluation check in the adverse-stress scenario shows near parity between DQN and the static-threshold baseline early in the horizon and a large, highly significant DQN advantage by the horizon midpoint, supporting a "locally tied, globally better" pattern. Episode-static behavior,

in which a controller fixes a threshold pair at the start of an episode and barely adjusts it, is no longer universal: it is specific to the DQN family at 12–30D and to PPO, while SAC and the actor-critic agents adjust thresholds within the episode at every dimension. We conclude that, given a sufficient training budget, RL comes within a few percent of well-designed rule-based weekly threshold search on profit, but the strongest rule-based policies still win the primary reward ranking in every scenario and dimension tested, and we find no robust risk-side advantage for RL.
Keywords: Reinforcement Learning, Credit Scoring, Threshold Control, Markov Decision Process, Delayed Rewards, Portfolio Simulation, Rule-Based Policies, State Dimensionality

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1. Introduction

A credit manager who holds a single static acceptance threshold loses money as the portfolio composition, the capital usage, and the market conditions drift between decisions. In practice, banks and consumer-lending institutions revise acceptance thresholds periodically rather than once. The fraction of applicants to approve this week depends on the current portfolio composition, outstanding capital commitments, rolling default rates, and market conditions that evolve between decisions. The weekly acceptance decision is therefore a recurring control problem, not a one-shot cut-off. Much of the academic literature has instead treated threshold selection as a one-shot classification problem, estimating a default probability, choosing a cut-off, and evaluating by AUC or Gini, which ignores the sequential nature of actual threshold management (Thomas et al., 2017; Lessmann et al., 2015). That

gap between static cut-off optimisation and dynamic portfolio management motivates the present study.

We build and evaluate an intelligent decision-support module for weekly threshold control, and we compare learned (RL) controllers against strong rule-based expert policies. Data-driven weekly threshold control is one concrete way artificial intelligence can improve business lending decisions. By adjusting acceptance thresholds to current portfolio and market signals, it can protect profit as conditions change, and the present study measures how close learned controllers come to well-designed rules and under which conditions. Reinforcement learning provides a natural framework for decisions with delayed consequences (Arulkumaran et al., 2017). An RL agent observes the state of the portfolio, selects a pair of weekly thresholds, and receives a reward signal that reflects realized portfolio cash flows rather than immediate cohort expectations. The delayed-outcome setting is central: a loan accepted at week t may not default or repay until week $t+\Delta$, so the correct evaluation object is the full portfolio trajectory, not the immediately observable cohort. Herasymovych et al. (2019) showed that a DQN agent can learn to adjust a single acceptance threshold in a simulated credit environment with delayed outcomes, establishing the structural ideas on which this study builds.

Calibration disclaimer. All experiments in this paper use a synthetic simulator calibrated to published retail lending statistics (Bank of Russia supervisory data and ECB SSM loan loss reports) as a controlled proof-of-concept test bench. Synthetic simulation is standard practice in RL-based credit research because real loan-level panels with weekly cohort granularity and observable repayment and default events are not publicly available (Herasy-

movych et al., 2019; Lessmann et al., 2015). The simulator provides full control over market regime, segment composition, and shock timing, which makes the multi-scenario robustness evaluation possible. Results should be interpreted as evidence about algorithmic behaviour under these controlled conditions, not as forecasts for any specific production portfolio.

The paper extends the framework of Herasymovych et al. (2019) with four contributions: (i) split weekly thresholds for new and repeat client segments, (ii) a controlled state-dimensionality experiment comparing 12D, 20D, 30D, and 50D observation sizes under a fixed protocol, (iii) five economically meaningful rule-based baselines as the comparison standard, and (iv) statistical hypothesis tests on paired run-level data. The central research question is:

> *Under which conditions does reinforcement learning provide a robust improvement over rule-based threshold policies in a delayed-outcome credit-scoring environment?*

The data give a largely negative answer. Rule-based policies win every scenario and dimension by the primary reward ranking (weekly grid search in five scenarios, the static threshold under adverse stress) and every scenario by raw profit. With a training budget large enough for the RL agents to approach convergence (1,500 episodes per seed) the best RL agent closes most of the profit gap to the leading baseline, to within 1.0–2.4% depending on the observation size, but it does not overtake the baseline anywhere. Nor does richer state produce a risk-side advantage: the best RL agent's default rate stays slightly above that of the leading baseline at every dimension, and it undercuts both profit-seeking baselines on default rate in at most two of six scenarios, while the risk-aware weekly policy stays the single lowest-default

controller overall. Observation size matters mainly for the value-based reference agent, whose reward peaks at an intermediate 20D–30D state and collapses at 50D.

2. Related Work

Traditional credit scoring estimates default probability at the application level and selects a cut-off that balances interest income against expected loss (Thomas et al., 2017; Hand & Henley, 1997; Siddiqi, 2006). Benchmarking studies have shown that classifier rankings depend on the dataset and the evaluation protocol, and that logistic regression and gradient-boosted trees are difficult to outperform with more complex classifiers (Lessmann et al., 2015; Baesens et al., 2003). The same conclusion recurs in more recent work: Gunnarsson et al. (2021) find that deep neural networks do not reliably beat gradient boosting on standard credit datasets, and reviews of consumer credit risk assessment reach a similar verdict on where modelling effort actually pays off (Zhang & Yu, 2023). Application-oriented research has therefore shifted from raw classifier accuracy toward feature enhancement and ensembling (Yang et al., 2024) and toward explainability for credit decisions (Nallakaruppan et al., 2024), reflecting the practical priorities of intelligent lending systems.

These advances still treat the accept/reject cut-off as a static one-shot tuning step. A second line of work reframes threshold choice as a cost-sensitive economic decision rather than a classification metric (Baesens et al., 2003), and Bellotti & Crook (2009) note that under Basel II/III capital adequacy requirements the threshold affects outstanding exposure, realized losses, and

regulatory capital in ways that make static cut-off selection insufficient once market conditions or portfolio composition drift over time. Once the decision is recognised as recurring and state-dependent, it becomes a sequential control problem, which is the natural domain of reinforcement learning.

Reinforcement learning addresses problems with delayed consequences through the Markov Decision Process (MDP) framework (Dayan & Niv, 2008; Arulkumaran et al., 2017). The deep RL algorithms used in this study illustrate the main design axes. DQN couples Q-learning with experience replay for off-policy sample efficiency on discrete action spaces, Double-DQN reduces the overestimation bias of vanilla DQN (van Hasselt et al., 2016), PPO is an on-policy policy-gradient method with stable updates (Schulman et al., 2017), SAC is an off-policy maximum-entropy actor-critic (Haarnoja et al., 2018), and A3C and A2C are asynchronous and synchronous actor-critic variants. A broad survey of these methods appears in Arulkumaran et al. (2017). In this study A2C, PPO, and SAC use the Stable-Baselines3 implementations (Raffin et al., 2021), while DQN, Double-DQN, and A3C use matched custom PyTorch implementations.

Applying RL to credit control is more recent and still thin. Herasymovych et al. (2019) optimise the acceptance threshold of a credit-scoring model with RL in a simulated environment with delayed outcomes, and Kiatsupaibul et al. (2024) survey reinforcement learning for credit scoring and underwriting and note how few controlled comparisons against strong rule-based policies exist. The direct precursor (Herasymovych et al., 2019) compared a DQN agent against a single static threshold in one scenario using a shared, non-split threshold. No prior work has simultaneously evaluated split thresholds

for distinct borrower segments, a controlled observation-dimensionality comparison, strong economically motivated rule-based baselines, and six market scenarios with statistical testing. That combination is the gap this paper fills, and it motivates the three hypotheses stated in Section 4.1.

3. Problem Formulation and Methodology

3.1. MDP Formulation

We model the weekly threshold-control problem as a discrete-time MDP $(\mathcal{S}, \mathcal{A}, \mathcal{P}, r, \gamma)$. The **state** $s_t \in \mathbb{R}^d$ ($d \in \{12, 20, 30, 50\}$) is a portfolio summary vector observed at the beginning of interactive week t. It includes approval rates, default rates, realized profit and NPV, capital usage, outstanding exposure, segment shares, and, at higher dimensionalities, lagged and rolling history. The **action** is a pair of acceptance thresholds $a_t = (\tau_t^{\text{new}}, \tau_t^{\text{repeat}}) \in \{35, 40, \ldots, 85\}^2$: applications from new (repeat) clients with credit score above τ_t^{new} (τ_t^{repeat}) are accepted. This yields $11 \times 11 = 121$ joint actions per week. The **reward** is described in Section 3.4. The discount factor is $\gamma = 0.98$.

Figure 1 shows the weekly decision loop as a Markov decision process. The figure makes explicit that each weekly action is scored only through delayed cash flows, so the controller must account for outcomes that arrive several weeks after the threshold is set.

3.2. Simulation Environment

Each episode consists of three phases. (1) An **8-week warm-up** prefills the loan book using default thresholds before the controller begins acting. (2) During the **52-week interactive horizon** the controller sets weekly

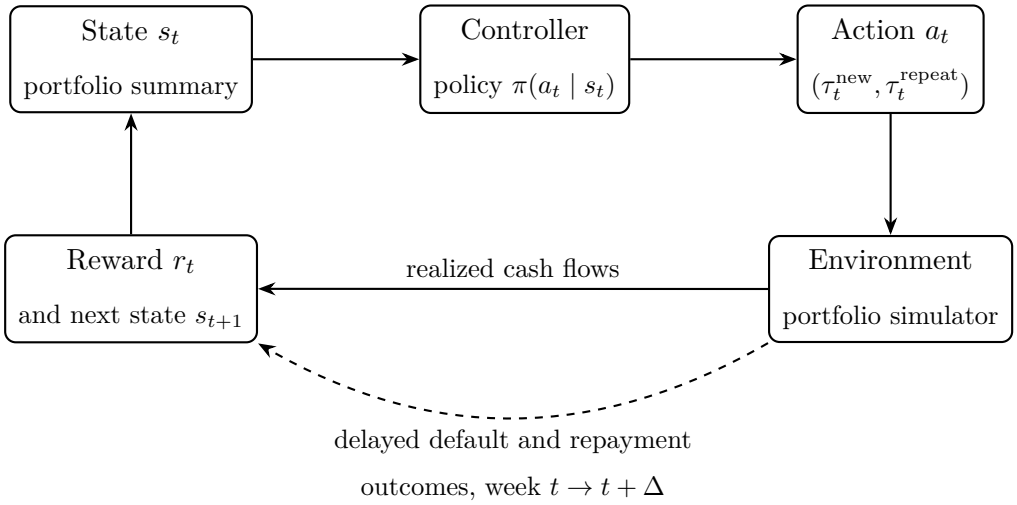


Figure 1: Weekly threshold control as a Markov decision process. The dashed arrow marks the delayed default and repayment outcomes that link an action taken at week t to rewards realized at later weeks $t + \Delta$.

thresholds. The simulator generates applications from Gaussian credit scores clipped to $[0, 100]$ for new and repeat segments, applies a logistic default-probability mapping, accepts loans above the threshold, and schedules repayment, default, and recovery events in a forward event queue. (3) **Terminal settlement** resolves all pending cash flows at episode end so that loans originated near the horizon are not mechanically penalized. Default probabilities are calibrated to target 4–6% in the base scenario and 9–14% under adverse stress, consistent with published retail lending statistics.

Figure 2 shows the three-phase episode structure and the within-week loan lifecycle. The figure shows that outcomes from accepted loans are pushed into a forward event queue, so repayments, defaults, and recoveries feed back into later weeks rather than into the week of origination.

In Table 1, the six evaluation scenarios and their primary stress mech-

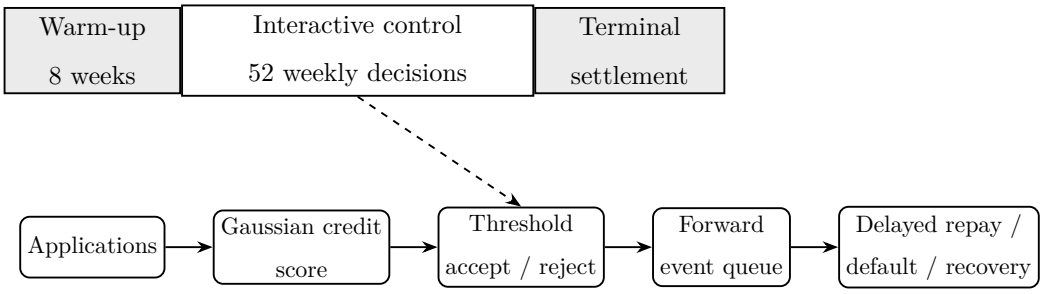


Figure 2: Episode structure and loan lifecycle. The top row is the episode timeline (8-week warm-up, 52-week interactive weekly control, terminal settlement), and the bottom row is the within-week sequence from application to delayed outcome.

anism are listed. The scenarios probe distinct portfolio-management weaknesses, so a controller that performs well across all of them is robust rather than tuned to a single regime.

3.3. Agents and Baselines

RL agents. All six agents use a shared two-layer MLP (2×128 hidden units) and a fixed set of hyperparameters held identical across the four state sizes (Appendix B). A2C, PPO, and SAC use Stable-Baselines3, while DQN, Double-DQN, and A3C use matched custom PyTorch implementations. No per-dimension hyperparameter search is performed, so any performance difference across dimensions is attributable to the observation vector rather than to the training recipe.

Rule-based baselines span five management styles. The static-threshold baseline (`static_threshold`) applies one fixed shared threshold. The split-static baseline (`split_policy_static`) maintains separate fixed defaults per segment. The profit-oriented policy (`profit_oriented`) performs a weekly grid search maximising current-cohort expected profit without penalty terms.

Table 1: The six evaluation scenarios and their primary stress mechanism.

Scenario	Stress mechanism
base_market	Stable, the cleanest test of policy signal exploitation
drift	Gradual new-client score deterioration and rising defaults
adverse_stress	Abrupt shock: higher defaults, larger loans, capital pressure
class_imbalance_shift	Growing share of weaker new applicants, tests composition risk
noise	Stable mean with noisy weekly volumes and measurements
split_policy_dynamics	New clients deteriorate while repeat clients improve

The risk-aware weekly policy (`risk_aware_weekly`) adjusts thresholds reactively to rolling default pressure and capital signals. The constraint-aware weekly policy (`constraint_aware_weekly`) runs a penalty-weighted weekly grid search that explicitly internalises default-rate, approval-rate, capital-usage, and volatility violations.

3.4. Reward Function

With delayed reward enabled, the shaped weekly reward is:

$$
\begin{aligned}
r_t = \; & w_p\, \Pi_t^{\text{realized}} + w_n\, \text{NPV}_t^{\text{realized}} - w_s\, \hat{d}_t \cdot 100 \\
& - \lambda_d \big[\max\big(0,\, \bar{d}_t - 0.12\big)\big]^2 \cdot 1000 - \lambda_a\, |\alpha_t - 0.42| \cdot 1000 \\
& - \lambda_c \big[\max\big(0,\, c_t - 0.82\big)\big]^2 \cdot 1000 - \lambda_v \max\left(0,\, \frac{\sigma_{\Pi,t} - 8500}{8500}\right) \cdot 1000,
\end{aligned}
\tag{1}
$$

where Π_t^{realized} and $\text{NPV}_t^{\text{realized}}$ are realized weekly profit and NPV (not current-cohort expectations), $\hat{d}_t$ is the expected default rate of the current cohort, $\bar{d}_t$ is the rolling realized default rate, α_t is the weekly approval rate, c_t is the projected capital usage ratio, and $\sigma_{\Pi,t}$ is realized profit volatility. The four constants are calibration targets: the default-rate cap (0.12), the target approval rate (0.42), the capital-usage cap (0.82), and the profit-volatility scale (8500). Active weights: w_p=1.0, w_n=0.25, w_s=0.35, λ_d=9.0, λ_a=2.0, λ_c=3.5, λ_v=1.4.

The reward is multi-objective by design. Rankings by cumulative reward need not coincide with rankings by raw profit, because the reward encodes portfolio risk constraints rather than profit alone. Throughout this paper, "realized profit" is computed from realized interactive weekly profits plus terminal settlement, that is, realized final portfolio profit rather than a current-cohort expectation.

4. Experimental Design

The experiment is fully controlled: *only* observation dimensionality varies across the four runs. Everything else (the simulator, the reward, the eight random seeds, the 1,500 training episodes per seed, the evaluation protocol, the scenarios, the threshold grid, and the controller set) is held fixed. Each of the six agents is trained once per state dimension per seed (6 agents $\times$ 4 dimensions $\times$ 8 seeds = 192 training runs, each running the full 1,500-episode budget with the training scenario drawn from a pool of all six market scenarios), and evaluated over 6 scenarios $\times$ 8 seeds $\times$ 12 evaluation runs = 576 evaluation runs per controller per dimension. A validity layer programmat-

ically verifies that the first 12 observation features are bit-for-bit identical across all four state sizes on a common reset and three fixed-action transitions.

In Table 2, the cumulative composition of the four observation layers is shown, where each layer adds features that a portfolio manager would realistically observe at weekly decision time with no access to future outcomes. Because each layer is a superset of the previous one, observation dimensionality is the only factor that changes across the four conditions.

4.1. Hypotheses

The three hypotheses follow from the research gap and from the literature reviewed in Section 2, and each is tested on paired run-level data in Section 5.4. The first hypothesis concerns state representation. Because feature enhancement and richer inputs are repeatedly reported to help credit models up to a point (Yang et al., 2024), we expect added observation features to improve the RL frontier but only over an intermediate range. The second hypothesis concerns the RL advantage. The credit-scoring benchmarking literature repeatedly finds that more complex models do not reliably dominate simpler ones (Lessmann et al., 2015; Gunnarsson et al., 2021), and the direct precursor compared RL against only a single static baseline (Herasymovych et al., 2019; Kiatsupaibul et al., 2024), so we expect the universal-dominance claim to fail against strong rule-based policies. The third hypothesis concerns delayed evaluation. Because outcomes are delayed and terminal settlement is applied, as in the delayed-reward credit-control setting of Herasymovych et al. (2019), early-period performance may misrepresent final portfolio quality.

Table 2: Cumulative observation layers. The first 12 features are invariant across all configurations and are verified programmatically before each experiment.

Size	Added information
12D	Compact dashboard: week progress, overall/segment approval rates, rolling realized default rate, expected default rate, realized profit, profit volatility, projected capital usage, outstanding ratio, both threshold levels.
20D	12D + current economic context: repeat segment share, expected profit/NPV per application, realized NPV, weekly reward, threshold gap, capital headroom, realized-vs-expected default gap.
30D	20D + lagged/rolling memory: second-lag approval/default/profit/capital, four-week rolling means and std of approval, default, and profit, lagged threshold changes.
50D	30D + segment-specific signals and stability descriptors: segment approval/ default/profit indicators, accepted-share ratios, rolling reward moments, cumulative reward/profit to date, capital-usage volatility, outstanding-ratio changes, threshold-gap history, application-volume ratio.

H1 (State representation).

Richer state representations improve the best RL frontier from 12D to 30D, while the move from 30D to 50D does not necessarily create additional improvement.

H2 (RL vs. rule-based).

RL improves over rule-based policies only in some scenarios and metrics, and the universal claim that RL robustly dominates strong rule-based

baselines is expected to be *rejected*.

H3 (Delayed evaluation).

Early-period performance is an unreliable proxy for final realized portfolio quality when outcomes are delayed and terminal settlement is applied. The "locally worse, globally better" pattern is expected to hold.

5. Results

5.1. Best Overall and Best RL Controller by Dimension

In Table 3, the strongest controller is reported when baselines and RL agents are pooled under the repository's official ranking rule (highest cumulative reward, then highest profit, then lowest default rate), together with the strongest RL controller in each dimension (full-profile run, 8 seeds $\times$ 1,500 episodes per seed, with data and configuration in the public repository). Unlike under the short 50-episode budget, the best RL agent is not the same algorithm at every dimension: SAC leads at 12D, 20D, and 50D, and A2C at 30D. The best RL profit is highest at 12D and declines steadily with observation size, but the gap to the leading baseline stays small at every dimension.

By cumulative reward, the primary ranking metric, the constraint-aware weekly policy is the best overall controller in every dimension (reward 3,666,052). On raw realized profit it is marginally below the profit-oriented policy (3,803,168 vs. 3,802,547). The two differ by less than 0.02%; over 576 paired runs the difference is statistically detectable ($p < 10^{-3}$) but not economically meaningful. The flat baseline row confirms that the experiment isolates the effect of observation size on RL quality, because the leading rule-based policy is inde-

Table 3: Best overall controller and best RL controller by state dimension (full-profile run, 1,500 training episodes per seed). The baseline line is flat because the leading baseline does not depend on the RL observation vector. "Profit" is realized final portfolio profit (Section 3.4); "Reward" is cumulative shaped reward.

Dim	Controller	Type	Profit	Reward	Default rate
Best overall (pooled, by cumulative reward)					
12	Constraint-aware weekly	baseline	3,802,547	3,666,052	0.0760
20	Constraint-aware weekly	baseline	3,802,547	3,666,052	0.0760
30	Constraint-aware weekly	baseline	3,802,547	3,666,052	0.0760
50	Constraint-aware weekly	baseline	3,802,547	3,666,052	0.0760
Best RL agent					
12	SAC	RL	3,766,009	3,622,122	0.0769
20	SAC	RL	3,750,108	3,628,740	0.0764
30	A2C	RL	3,719,303	3,602,141	0.0766
50	SAC	RL	3,710,014	3,585,105	0.0767

pendent of the RL state vector by construction. With the 1,500-episode budget, the profit gap between the best RL agent and the best overall controller shrinks to −36,538 (−0.96%) at 12D, −52,439 (−1.38%) at 20D, −83,244 (−2.19%) at 30D, and −92,534 (−2.43%) at 50D, and the corresponding reward gaps are −43,930 (−1.20%), −37,313 (−1.02%), −63,911 (−1.74%), and −80,947 (−2.21%). Every gap is under 2.5% of the baseline value, compared with −9.4% for the best RL agent's 12D peak under the short 50-episode budget, so the dominant effect of the larger training budget is to close most of the RL-to-baseline gap rather than to change which controller wins. Default rate for the best RL agent is 7.69% at 12D, 7.64% at 20D, 7.66% at

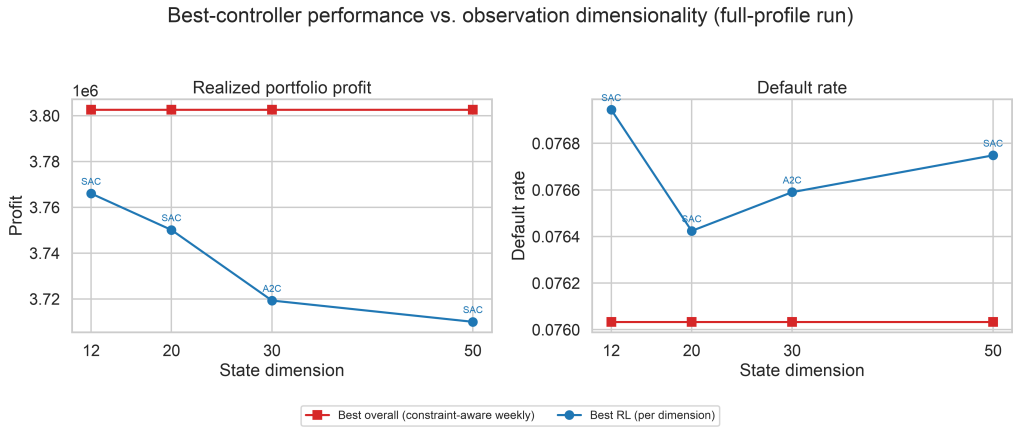


Figure 3: Best-controller performance versus observation dimensionality (full-profile run, 8 seeds, 1,500 training episodes per seed). *Left*: realized portfolio profit. *Right*: default rate. The best overall controller (the constraint-aware weekly policy) is independent of the RL state vector and is therefore flat across dimensions, while the best RL agent (labelled per point: SAC at 12D, 20D, and 50D, A2C at 30D) sits close to the baseline on profit, with a gap that widens with dimension, and slightly above the baseline on default rate at every dimension.

30D, and 7.67% at 50D, slightly above the leading baseline's 7.60% at every dimension.

Figure 3 shows the same comparison as a function of observation dimensionality. The figure shows that the best RL agent's profit is close to the leading baseline at every dimension but drifts further below it as the observation grows, while on default rate the best RL agent stays above the baseline line at every dimension, by between 0.04 and 0.09 percentage points. Richer state therefore changes neither who wins the reward ranking nor the sign of RL's default-rate gap.

Table 4: Agent and baseline summary (mean across all four dimensions, full-profile run, 1,500 training episodes per seed). "Approval" denotes mean weekly approval rate. Baseline rows do not depend on state dimension, so their per-dimension values are identical and the mean equals any single dimension's value.

Controller	Type	Mean profit	Default rate	Approval
Profit-oriented	baseline	3,803,168	7.61%	94.91%
Constraint-aware weekly	baseline	3,802,547	7.60%	94.84%
Static-threshold	baseline	3,083,370	6.62%	69.99%
SAC	RL	3,702,324	7.65%	92.69%
A2C	RL	3,607,319	7.59%	90.12%
A3C	RL	3,320,193	7.02%	78.48%
DQN	RL	3,262,483	7.15%	79.18%
Double-DQN	RL	3,131,792	7.16%	74.99%
PPO	RL	2,617,957	6.21%	57.16%

5.2. Agent Comparison

In Table 4, the mean performance of each agent across all four dimensions is summarised alongside the three principal rule-based baselines (full-profile run, 1,500 training episodes per seed). With this larger budget the RL ranking is very different from a short-budget run: SAC leads all RL agents on mean profit and comes within 2.7% of the best baseline's mean profit, PPO remains the weakest and most degenerate agent, and the DQN family is not the strongest RL agent on average even though it supplies the canonical reference agent for the hypothesis tests in Section 5.4.

SAC leads all RL agents on mean profit (3,702,324), only 2.7% below the leading baseline's mean profit (3,803,168), a striking change from the -9.4%

gap of the best RL agent under a short 50-episode budget. A2C is second (3,607,319). DQN and Double-DQN, the two off-policy value-based agents, are in the lower half of the table on mean profit (3,262,483 and 3,131,792), mostly because of a collapse at 50D, where DQN's profit falls to 2,389,264 from 3,597,632 at 30D; DQN is not the best RL agent at any dimension (Table 3). PPO remains the weakest and most degenerate RL agent: its profit, default rate, and approval rate are numerically identical across 12D, 20D, 30D, and 50D (2,617,957 profit at every dimension), which means PPO converges to the same policy regardless of the additional observation features even under the larger training budget, so those features remain unused rather than uninformative. Between the two extremes, mean default rate tracks mean approval rate across agents (higher-approval agents accept more marginal applicants and so post higher default rates), which is consistent with a credit-quality trade-off along the approval axis rather than a free improvement from any single agent.

5.3. Scenario-Level Winners

In Table 5, the reward-based, profit-based, and lowest-default-rate winner for each scenario is reported; the winners are identical at all four dimensions. No single metric identifies one controller as best, because the reward, profit, and default-rate winners differ across scenarios.

RL wins zero of six scenarios by reward and zero of six by profit at every dimension. The profit-oriented policy wins raw profit in five scenarios and the constraint-aware weekly policy in adverse_stress, while the reward winner is the profit-oriented, constraint-aware weekly, or static-threshold policy depending on the scenario, which is why the headline "best overall" ranking,

Table 5: Scenario-level winners (full-profile run, 1,500 training episodes per seed; all three columns are identical at 12D, 20D, 30D, and 50D, and every winner is a rule-based policy). *Reward winner*: highest cumulative shaped reward. *Profit winner*: highest realized profit. *Default winner*: lowest default rate. The risk-aware weekly policy is the single lowest-default controller in every scenario at every dimension.

Scenario	Reward winner	Profit winner	Default winner
adverse_stress	Static-threshold	Constraint-aware weekly	Risk-aware weekly
base_market	Profit-oriented	Profit-oriented	Risk-aware weekly
class_imbalance_shift	Constraint-aware weekly	Profit-oriented	Risk-aware weekly
drift	Profit-oriented	Profit-oriented	Risk-aware weekly
noise	Profit-oriented	Profit-oriented	Risk-aware weekly
split_policy_dynamics	Constraint-aware weekly	Profit-oriented	Risk-aware weekly

based on cumulative reward, and the profit ranking need not coincide.

The best RL agent also shows no systematic default-rate advantage over the two profit-seeking baselines (profit-oriented and constraint-aware weekly). Counting per-scenario wins against *both* profit-seeking baselines simultaneously, the best RL agent undercuts them in 0 of 6 scenarios at 12D (SAC), 2 of 6 at 20D (SAC, in the noise and split_policy_dynamics scenarios), 1 of 6 at 30D (A2C, noise), and 1 of 6 at 50D (SAC, noise). The only recurring case is the noise scenario, and the count does not grow with observation size, so richer state does not turn RL into a lower-risk alternative to the profit-seeking baselines. The risk-aware weekly policy remains the single lowest-default controller in every scenario at every dimension (Table 5), at the cost of far lower approval and profit. For a portfolio manager who values capital conservation, the relevant comparison is therefore between rule-based

policies with different risk settings, not between RL and rules.

5.4. Statistical Hypothesis Tests

In Table 6, the paired statistical comparisons for the three hypotheses of Section 4.1 are summarised, all computed on the full-profile run (1,500 training episodes per seed), with the paired run-level and weekly-trajectory data in the public repository. The tests show that the reference agent's reward peaks at an intermediate state (20D ahead of 30D, 30D ahead of 12D and far ahead of 50D), that RL (represented by the reference DQN agent) is significantly weaker than the profit-oriented baseline on reward at 30D and 50D, and that the RL deficit under stress reverses well before the end of the horizon. The dimensionality hypotheses are tested on vanilla **DQN**, used here as the canonical discrete-action reference agent for the controlled state-size comparison; DQN is not the best RL agent at any dimension (Table 3) or on average (Table 4, where SAC leads), so DQN is used consistently as the fixed reference rather than as a claim that it is the strongest agent. Each paired observation matches the two controllers on training seed, scenario, and evaluation run index. In the H1 comparisons both sides are RL agents evaluated on identical environment seeds, so each difference is taken on the same simulated market draws. In the H2 and H3 comparisons the rule-based baseline is evaluated on environment seeds offset by a fixed constant from the agent's, so these pairs are matched by index rather than on common random numbers, and the pairing does not remove market-draw noise from these differences.

H1 (richer state improves RL up to an intermediate representation) is *partially supported*: the reference agent gains from a richer state up to an

intermediate size and loses heavily beyond it, but the optimum lies at 20D rather than 30D. The 30D to 12D reward difference is positive and highly significant ($p < 10^{-3}$, mean +54,056, $n = 576$ paired runs), so 30D reliably beats 12D under the larger training budget. The 30D to 20D difference, however, is significantly *negative* ($-71{,}104$, $p < 10^{-3}$), so the lagged and rolling memory added at 30D does not improve on the 20D economic context for this agent. The 30D to 50D comparison is highly significant by a wide margin (+976,844, $p < 10^{-3}$), which confirms that 50D reliably underperforms the intermediate states. A supplementary check against 20D Double-DQN (the other value-based agent) gives a mixed picture: 30D DQN beats 20D Double-DQN on reward (+395,097, $p < 10^{-3}$) but has a significantly *higher* default rate by 0.33 percentage points ($p < 10^{-3}$). On the best-RL frontier (Table 3), H1 is not supported: the best RL profit declines from 12D to 50D and the best RL reward is nearly flat.

H2 (RL does not universally dominate) remains confirmed. The 30D DQN agent is significantly weaker than the profit-oriented policy on reward, by $-121{,}769$ ($p < 10^{-3}$, a 3.3% gap), down from $-368{,}145$ under the short-budget run. The 50D DQN agent trails by $-1{,}098{,}613$ ($p < 10^{-3}$), a large gap that reflects DQN's collapse at 50D (Table 4) rather than a property of RL at 50D in general, since SAC is the best RL agent at 50D and comes much closer to the baseline (Table 3). 30D DQN has a significantly *lower* default rate than the profit-oriented policy (-0.24 percentage points, $p < 10^{-3}$), but it buys this with a lower approval rate (88.6% versus 94.9%) and lower profit (3,597,632 versus 3,803,168), so it is a move along the approval–risk trade-off rather than a free risk-side improvement.

H3 (delayed evaluation changes rankings, "locally worse globally better") is *supported*, and more strongly than under the short-budget run. The H3 checkpoints are computed directly on the full-profile dim-30 run's weekly trajectories in the adverse_stress scenario ($n = 96$ paired observations at each checkpoint), not on a separate development profile. At the early checkpoint (interactive week 6) DQN is statistically indistinguishable from the static-threshold baseline (profit gap $-1{,}032$, $p = 0.414$), essentially tied rather than trailing. By week 25, roughly the midpoint of the 52-week interactive horizon, DQN has opened a large and highly significant advantage over the static baseline (profit gap $+124{,}995$, $p < 10^{-3}$). The pattern is therefore "locally tied, globally better" rather than the persistent deficit found under the short-budget run: the RL agent's advantage under stress accumulates over the horizon as delayed outcomes settle, and it is already decisive well before the horizon ends, which argues against relying on early-horizon snapshots to judge RL policies under delayed-outcome evaluation.

6. Discussion

6.1. Why Rule-Based Baselines Still Win

The leading baselines (the constraint-aware weekly and profit-oriented policies) perform a weekly grid search over the same 121 joint threshold actions as the RL agents. Their advantage is structural: they evaluate each action directly on the current week's cohort economics, they see the realized profit and default signals immediately, and they set thresholds according to an explicit objective, so they need no training at all. Raising the training budget from 50 to 1,500 episodes per seed closes most, but not all, of the

gap: the best RL agent's profit shortfall shrinks from −9.4% to between −1.0% and −2.4% depending on dimension (Table 3), and the reference DQN agent's reward deficit against the profit-oriented policy at 30D falls from −368,145 to −121,769 (Section 5.4). This shows that training budget, not some fundamental ceiling on RL, was the dominant factor behind the earlier short-budget gap. What remains after 30× more training is a much smaller residual gap that has the same sign in every scenario and dimension tested and is significant when pooled over scenarios, although in several individual scenarios at 12D and 20D it is no longer statistically significant. This is consistent with a genuine structural advantage for direct weekly evaluation over a learned value function under delayed, noisy rewards, rather than an artifact of undertraining.

6.2. No Robust Risk-Side Advantage for RL

A natural hope is that RL, even when it trails on profit, buys a lower default rate. The data do not support this. The best RL agent's mean default rate is slightly above that of the leading baseline at every dimension (Table 3), and counting scenarios in which the best RL agent beats both the profit-oriented and constraint-aware weekly policies on default rate simultaneously gives 0 of 6 at 12D, 2 of 6 at 20D, and 1 of 6 at both 30D and 50D (Section 5.3), almost always in the noise scenario. Where an RL agent does post a lower default rate, as 30D DQN does against the profit-oriented policy (Section 5.4), it approves fewer applicants and earns less, which is the same approval–risk trade-off that the rule-based policies already span: the risk-aware weekly policy remains the lowest-default controller in every scenario at every dimension (Table 5). A portfolio manager who values capital con-

servation can therefore obtain it more cheaply by tuning a rule-based policy than by deploying RL in this setting.

6.3. Sensitivity of the RL Ranking

The ranking among the leading RL agents within a dimension is not robust. Across the eight training seeds, the standard deviation of a best RL agent's per-seed mean profit is between 59,359 (SAC, 20D) and 192,933 (A2C, 30D), against 12,113 for the constraint-aware weekly baseline, whose per-seed variation reflects only evaluation noise. The margins that decide the best RL agent are often smaller than this spread: at 20D, A2C trails SAC by 4,170 in reward but leads it by 11,453 in profit. The identity of the best RL agent at a given dimension should therefore be read as indicative, whereas the conclusion that RL trails the best rule-based policy holds in every scenario at every dimension.

6.4. Dimensionality Saturation

The 30D state adds lagged and rolling history (second-lag approval, default, and profit values, four-week rolling means and standard deviations, and lagged threshold changes). In principle this temporal context helps under slow-moving drift or noisy volume signals, because it lets a controller compare this week with the recent past. The 50D state adds richer segmentation and stability descriptors, but those extra features require more optimization work and offer diminishing marginal information at the current training budget. In practice, for the reference DQN agent the reward optimum lies at 20D: 20D beats 30D, 30D beats 12D, and 50D falls far behind all three (Section 5.4). The 20D layer adds the current economic context

(expected profit and NPV per application, capital headroom, the realized-versus-expected default gap), which bears directly on the weekly decision, whereas the lagged and rolling memory added at 30D brings no further gain for this agent, and the segment and stability descriptors added at 50D make the value function markedly harder to learn at the same budget. Saturation is algorithm-dependent, however: the best-per-dimension profit declines steadily from 12D to 50D, SAC remains competitive at every size, and A2C peaks at 20D–30D (Table 3), so there is no single dimension at which every algorithm peaks. The clearest effect is at the top of the range: from 30D to 50D the DQN family loses heavily and A2C loses moderately, while SAC and A3C are unaffected or gain slightly, so the 50D features help only the agents that already handle larger inputs well.

6.5. Episode-Static RL Behavior

Under the short 50-episode budget, every best-per-dimension RL agent had threshold volatility of 0.00, and the finding was that RL policies are uniformly episode-static. Under the larger 1,500-episode budget this is no longer true: all four best-per-dimension agents show real within-episode threshold adjustment (SAC at 12D, volatility 2.05; SAC at 20D, 3.47; A2C at 30D, 1.46; SAC at 50D, 4.72). Looking at all six agents across all four dimensions, the pattern is algorithm-specific rather than dimension-specific: DQN and Double-DQN (the two off-policy, greedy value-based agents) stay at 0.00 threshold volatility at 12D, 20D, and 30D and only begin adjusting thresholds within the episode at 50D (volatility 3.81 and 2.77 respectively); PPO remains at 0.00 at every dimension, consistent with its degenerate, dimension-invariant policy (Section 5.2); while SAC and the two

actor-critic agents (A2C, A3C) show nonzero within-episode adjustment at every dimension, increasing with state size for SAC (2.05 at 12D up to 4.72 at 50D). Episode-static behavior is therefore a property of the DQN family's greedy, off-policy exploitation at low-to-moderate dimensionality, not a general RL phenomenon: entropy-regularized and on-policy actor-critic agents adapt within the episode throughout, and even the DQN family does so once the state is large enough. The implication for practitioners is that the choice of algorithm, not just training budget or state size, determines whether a controller performs genuine week-to-week adaptation or effectively episode-level threshold selection.

6.6. Practical Recommendations

For a lending business the results translate into concrete guidance.

- Training budget matters more than the choice between RL and rule-based search: moving from 50 to 1,500 training episodes per seed cut the best RL agent's profit shortfall from -9.4% to under -2.5% at every dimension, so an institution considering RL should budget for substantial training before judging the approach on results from a short pilot run.
- Rule-based policies still win the primary reward ranking in every scenario and dimension (the constraint-aware or profit-oriented weekly grid search in five scenarios, the static threshold under adverse stress) and need no training, so they remain the safe default when training resources or validation time are limited.
- Do not expect RL to lower the default rate for free: in this setting the best RL agent's default rate is slightly above the leading baseline's at

every dimension, and an institution that values default-rate reduction can obtain it more directly by tuning the risk settings of a rule-based weekly policy.

- If RL is deployed, prefer a compact or intermediate state (12D–30D) and an agent that handles it robustly (SAC or A2C here). Larger observation vectors bring no profit gain at this training budget and can severely degrade value-based agents (DQN at 50D), and because the best agent at a given size varies with the training seed, several seeds should be trained and compared before choosing one.

6.7. Limitations

First, all results derive from a synthetic simulator, so calibration to any specific production portfolio would shift the absolute numbers and potentially alter some rankings. Second, although the 1,500-episode-per-seed training budget is 30× larger than the short-budget run and closes most of the RL-to-baseline gap, it is not a proof of full convergence; a still larger budget could narrow the residual gap further, particularly for the DQN family at 50D. Third, controller rankings are sensitive to the reward weights in Equation (1), and a reward sensitivity analysis is left for future work. Fourth, bootstrap confidence intervals quantify simulation uncertainty, not real-world external validity. Fifth, the H3 delayed-evaluation check is reported for a single scenario (adverse_stress) and a single dimension (30D); whether the same "locally tied, globally better" pattern holds in other scenarios and dimensions is left for future work. Sixth, eight training seeds are enough to separate RL from the leading baselines but not to rank the leading RL agents against each other reliably (Section 6.3), and the RL results depend on network initiali-

sation and on the deep-learning library version, which is why the repository records the exact package versions used.

These results are bounded by the synthetic setting. The default and recovery parameters are calibrated to aggregate statistics rather than fitted to a single proprietary book, so the absolute profit and default levels should be read as order-of-magnitude rather than exact. Even at 1,500 training episodes per seed, a still larger budget could narrow the residual RL-to-baseline gap further, so the comparison in this paper should be read as "RL under a substantial, but not unlimited, training budget" rather than as an asymptotic comparison. The real-data comparison validates the score-to-default relationship and the parameter ranges, but it does not validate the weekly threshold dynamics directly, which would require a lender to deploy and log a controlled threshold policy.

7. Threats to validity

The conclusions of this study rest on a synthetic simulator, so their external validity depends on how faithfully the simulator reproduces the dynamics of real unsecured consumer lending. This section states the main threats, compares the simulator against published real-world evidence, and explains why an agent-based simulator remains the appropriate tool under these threats.

7.1. Construct validity and simulator fidelity

The portfolio is generated from parametric assumptions rather than observed loans, which introduces three concrete risks. First, the logistic default model in Appendix A imposes a smooth monotone link between score and

default probability, whereas real default curves can be locally non-monotone in thin score regions. Second, the weekly event schedule collapses repayment, default, and recovery into a small set of timed cashflows, which understates the heterogeneity of real repayment paths. Third, the scenario set spans six stylised regimes, so it cannot represent every macro-financial path a real book would face. We mitigate these risks by calibrating every primary parameter to published statistics and by comparing the score-to-default relationship against a large real loan book, as described next.

7.2. Comparison with published evidence and a real loan book

Real consumer credit portfolios show a wide spread of default rates and strong class imbalance (Brown & Mues, 2012), which the scenario set is designed to span. Table 7 places each primary simulator parameter next to the corresponding real-world range and its source. In Table 7, the portfolio default rate, the loan tenor, the effective interest, and the recovery assumption all fall inside the ranges reported for unsecured retail lending, and the acceptance-threshold mechanism reproduces the setup of Herasymovych et al. (2019), whose optimal acceptance threshold on a real online consumer-loan book sits in the same mid-score region used here. Figure 4 overlays the simulator default-probability curves on the empirical default rate by within-book credit-score percentile estimated from the Lending Club loan book (Xia et al., 2017). In Figure 4, plotting against the within-book percentile aligns the two score domains, and both relationships fall monotonically while the higher Lending Club levels reflect its higher-rate and longer-tenor borrower mix.

Table 7: Calibration of the simulator against published evidence. In Table 7, each primary parameter sits inside a real-world range, and the sources are aggregate statistics and peer-reviewed studies rather than a single proprietary book.

Parameter	Simulator	Real-world range (source)
Portfolio default rate	4–14% across scenarios	around 4–5% in normal conditions and up to 12.9% under stress for unsecured consumer loans (Bank of Russia, 2025) 20.0% on Lending Club resolved loans (own computation; Xia et al., 2017)
Acceptance mechanism	weekly score threshold, grid 35–85	optimal threshold in the mid-score region on a real online book (Herasymovych et al., 2019)
Loan tenor	4–12 months	short-term consumer and online installment products (Herasymovych et al., 2019) Lending Club is longer, with 76% at 36 months and 24% at 60 months
Loan amount	0.3–9.0 k currency units	small-ticket loans typical of short-term online consumer lending (Herasymovych et al., 2019) Lending Club median is larger, at 12,000 USD
Effective interest	11–15% annualised	comparable to the Lending Club median of 12.7%
Recovery on default	26–34% (LGD $\approx$ 66–74%)	recovery distributions for unsecured consumer loans (Loterman et al., 2012; Qi & Zhao, 2011)

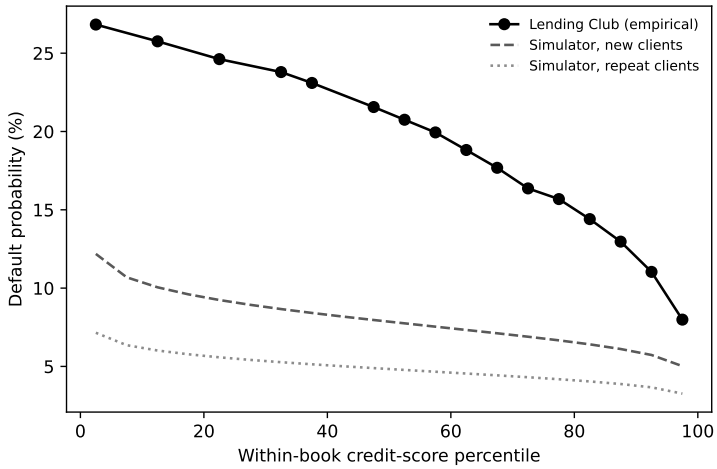


Figure 4: Default probability by within-book credit-score percentile, simulator curves versus the Lending Club loan book. In Figure 4, the within-book percentile axis aligns the two score domains, and the higher Lending Club levels reflect its higher-rate and longer-tenor borrower mix.

7.3. Why an agent-based simulator

Loan-level data that jointly record the score, the funding decision, the realised outcome, and the calendar timing across several market regimes are rarely public, because lender underwriting models and outcome histories are proprietary assets (Di Maggio et al., 2024). In the absence of such data, an agent-based simulator acts as a controlled laboratory that varies one factor at a time, here the observation dimension, while holding the protocol fixed, which is the comparison that observational data cannot deliver (Axtell & Farmer, 2025). We therefore read the simulator as a tool for studying the structure of the control problem under a fixed training budget, not as a forecast of any single lender's profit.

8. Conclusion

This paper evaluates six reinforcement learning algorithms against five rule-based baselines across six market scenarios and four observation-space sizes (12D, 20D, 30D, 50D) in a controlled credit portfolio simulation, at a training budget of 1,500 episodes per seed. The main empirical findings are:

(1) The constraint-aware weekly policy remains the best overall controller in every dimension by cumulative reward (reward 3,666,052). On raw realized profit it is within 0.02% of the profit-oriented policy (3,802,547 vs. 3,803,168, the latter marginally higher). RL wins zero of six scenarios by reward and zero of six by profit at every dimension.

(2) The best RL agent is not a single algorithm: SAC leads at 12D, 20D, and 50D (profit 3,766,009, 3,750,108, and 3,710,014), and A2C leads at 30D (3,719,303). The profit gap to the best overall controller shrinks to between -0.96% (12D) and -2.43% (50D), widening with dimension, down from -9.4% for the best RL agent (Double-DQN, 12D) under a $30\times$ smaller training budget. The margins between the leading RL agents are smaller than their seed-to-seed spread, so this ranking is indicative only.

(3) For the reference DQN agent, reward peaks at an intermediate state: 30D beats 12D ($+54{,}056$, $p < 10^{-3}$), 20D beats 30D ($+71{,}104$, $p < 10^{-3}$), and 50D falls far behind ($-976{,}844$ against 30D, $p < 10^{-3}$). H1 is therefore partially supported, with the optimum at 20D rather than 30D.

(4) RL shows no robust risk-side advantage: the best RL agent's default rate is slightly above the leading baseline's at every dimension, and it undercuts both profit-seeking baselines in at most two of six scenarios

(0, 2, 1, and 1 of 6 at 12D, 20D, 30D, and 50D); the risk-aware weekly policy stays the lowest-default controller overall at every dimension.

(5) A delayed-evaluation check in the adverse-stress scenario shows DQN statistically tied with the static-threshold baseline early in the horizon (week 6, $p = 0.414$) and significantly ahead by the horizon midpoint (week 25, +124,995, $p < 10^{-3}$), a "locally tied, globally better" pattern, the opposite of the persistent early-and-late deficit found under the short-budget run.

(6) Episode-static behavior, in which threshold volatility is 0.00 throughout the episode, is no longer universal: it is specific to the DQN family at 12–30D (which only begins adjusting thresholds within the episode at 50D) and to PPO at every dimension, while SAC and the actor-critic agents (A2C, A3C) adjust thresholds within the episode at every dimension tested.

The practical implication has shifted from the short-budget run: training budget, not a fundamental RL-versus-rule-based ceiling, was the dominant factor behind the earlier large gap. Rule-based policies still win the primary reward ranking in every scenario and dimension tested, and the weekly grid search needs no training, so it remains the safe default when training resources are limited. A sufficiently trained RL agent comes within a few percent of it on profit, but it does not overtake it in any scenario and does not offer a lower default rate in exchange, so in this setting RL is a close second rather than a better alternative. A compact or intermediate observation (12D–30D) is the sensible choice for an RL deployment, since larger states bring no profit gain at this budget and can severely degrade value-based agents. Future work

should investigate whether an even larger training budget closes the remaining -0.96% to -2.43% profit gap entirely (particularly for the DQN family at 50D), how many seeds are needed to rank the leading RL agents reliably, reward sensitivity to penalty weights, more adaptive recurrent or attention-based architectures, whether the "locally tied, globally better" pattern generalizes beyond the adverse-stress/30D pairing tested here, and validation on real or semi-synthetic loan panels.

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Appendix A. Simulator details

This appendix specifies the data-generating process so that the experiments can be reproduced from the public repository. All values are the calibrated full-profile settings.

Applicant generation.. Each interactive week the simulator draws a base application volume around 600 applications, scales it by a seasonal factor and multiplicative volume noise, and splits it into new and repeat counts through two Poisson draws. The repeat share follows a scenario base level modulated by an annual seasonal term and clamped to $[0.15, 0.75]$. Each applicant in segment s receives a credit score drawn from a Gaussian $\mathcal{N}(\mu_s, \sigma_s)$ clipped to $[0, 100]$, with $\mu_{\text{new}} = 59$, $\sigma_{\text{new}} = 13$ and $\mu_{\text{repeat}} = 69$, $\sigma_{\text{repeat}} = 10$ in the base market.

Default and recovery.. An applicant with score x in segment s defaults with probability

$$p_s(x) = \sigma\big(\beta_s - \kappa_s \cdot \tfrac{x-50}{10}\big)\,, \tag{A.1}$$

clipped to $[0.01, 0.98]$, where σ is the logistic function and $\beta_{\text{new}} = -2.3$, $\kappa_{\text{new}} = 0.18$, $\beta_{\text{repeat}} = -2.6$, $\kappa_{\text{repeat}} = 0.20$ in the base market. Scenario

stress shifts β_s gradually over the horizon and an abrupt shock adds to β_s from the shock week onward (Table A.8). These settings target portfolio default rates of 4–6% in the base market and 9–14% under adverse stress. A defaulted loan recovers with probability $r_{\text{new}} = 0.26$ or $r_{\text{repeat}} = 0.34$, and on recovery the lender receives the principal plus a fraction φ_s of the scheduled interest, with $\varphi_{\text{new}} = 0.65$ and $\varphi_{\text{repeat}} = 0.72$.

Loan terms and cashflows.. The principal is lognormal with mean $A_{\text{new}} = 1650$ or $A_{\text{repeat}} = 2150$ currency units and log-scale 0.28 (new) or 0.25 (repeat), clipped to $[300, 9000]$ and scaled by the scenario loan-amount multiplier. Each loan receives an integer term d in months (new: 5–12, repeat: 4–10) and a scheduled interest income of principal $\times$ $(\rho_s\, d/12 + o_s)$, with annual fee rate $\rho_{\text{new}} = 0.16$, $\rho_{\text{repeat}} = 0.13$ and origination $o_{\text{new}} = 0.035$, $o_{\text{repeat}} = 0.03$. Outcomes enter a forward event queue: a repaid loan returns the interest at week issue $+ d$, a defaulted loan returns $-$principal at week issue $+ d$, and a recovered loan adds the recovery value a further 1–4 weeks later. The term index d therefore sets both the interest horizon $d/12$ years and the event delay d weeks. Each cashflow is discounted to net present value at a weekly rate of $0.14/52$, that is an annual discount rate of 14%.

Episode structure and capital.. Each episode runs an 8-week warm-up at the default thresholds (60 for new, 50 for repeat), then a 52-week interactive horizon, then a terminal settlement that resolves all pending events so that loans originated near the horizon are not penalised. Outstanding principal plus the principal accepted in the current week is tracked against a capital limit of 3,000,000 to form the capital-usage ratio used in the reward.

Table A.8: Per-scenario stress parameters. In Table A.8, the score and default shifts are total drifts applied gradually over the horizon, and the shock column gives the magnitude added to the default intercept and the week it begins.

Scenario	Repeat share	Score shift (new/rep)	Default shift (new/rep)	Score noise	Loan mult.	Shock (week)
base_market	0.35	0.00 / 0.00	0.00 / 0.00	1.00	1.00	none
drift	0.34	−0.18 / −0.07	0.18 / 0.08	1.00	1.00	0.08 (13)
adverse_stress	0.33	−0.10 / −0.10	0.30 / 0.22	1.15	1.15	0.22 (10)
class_imbalance_shift	0.28	−0.08 / 0.03	0.12 / −0.02	1.00	0.95	0.12 (16)
noise	0.35	0.00 / 0.00	0.04 / 0.04	1.45	1.00	none
split_policy_dynamics	0.38	−0.20 / 0.10	0.22 / −0.10	1.00	1.00	0.06 (14)

Appendix B. Training hyperparameters

In Table B.9, the shared training configuration for the full-profile run is listed. The settings are held identical across all four state sizes, so any performance difference between dimensions is attributable to the observation vector rather than to the training recipe.

Table B.9: Shared training settings (full-profile run), held identical across all four state sizes. In Table B.9, these settings are fixed so that any difference across dimensions reflects the observation vector rather than the training recipe.

Parameter	Value	Notes
Network architecture	MLP, 2×128 hidden units	Shared across all agents
Discount factor γ	0.98	Standard
Learning rate	SB3 default (varies by agent)	Not tuned
Training episodes/seed	1,500	Full profile
Seeds	11, 23, 37, 51, 67, 79, 97, 113	8 seeds, fixed across dims
Evaluation runs	12 per seed per scenario	Same protocol across dims
Threshold grid	$[35, 85]$ in steps of 5	11 actions per segment
Warm-up weeks	8	Pre-fills loan book
Interactive horizon	52 weeks	Full profile
Applications per week	600	Before scale factors
Bootstrap resamples	1000	95% CI

Table B.10: Per-agent reinforcement-learning hyperparameters (full-profile run), held identical across all four state sizes. In Table B.10, A2C, PPO, and SAC use Stable-Baselines3 (Raffin et al., 2021), while DQN, Double-DQN, and A3C use matched custom PyTorch implementations.

Agent	Implementation	Key hyperparameters
DQN	Custom	lr 10^{-3}, batch 64, replay 2×10^4, learning starts 300, train freq 1, gradient steps 1, target update 200, ε $1.0 \rightarrow 0.05$ over 65% of training
Double-DQN	Custom	As DQN, with a double-Q target
PPO	SB3	lr 3×10^{-4}, rollout $n = 256$, batch 64, GAE λ 0.95, entropy coef 0.01, clip range 0.20
A2C	SB3	lr 7×10^{-4}, rollout $n = 256$, GAE λ 0.95, entropy coef 0.01
A3C	Custom	lr 5×10^{-4}, entropy coef 0.01, value coef 0.5, full-episode rollouts (single worker)
SAC	SB3	lr 3×10^{-4}, buffer 5×10^4, learning starts 500, batch 128, τ 0.02, train freq 1, gradient steps 1

CRediT authorship contribution statement

Evgeny Baulin: Software, Investigation, Writing – original draft. **P. P. Lukianchenko:** Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Table 6: Paired statistical checks (full-profile run, 1,500 training episodes per seed). The H3 rows use the dim-30 weekly trajectories in the adverse_stress scenario; week 25 is roughly the midpoint of the 52-week interactive horizon (interactive weeks are indexed 0–51), not the final week. *Mean diff.*: left minus right controller, where positive values favour the left controller on reward/profit and negative values favour it on default rate. CI: 95% percentile bootstrap.

Hyp.	Paired comparison	Metric	Mean diff.	95% CI	p
H1	30D DQN − 12D DQN	Reward	+54,056	[+33,245, +74,534]	$<10^{-3}$
H1	30D DQN − 20D DQN	Reward	−71,104	[−98,737, −44,861]	$<10^{-3}$
H1	30D DQN − 50D DQN	Reward	+976,844	[+892,740, +1,065,462]	$<10^{-3}$
H2	30D DQN − Profit-oriented	Reward	−121,769	[−150,029, −95,145]	$<10^{-3}$
H2	50D DQN − Profit-oriented	Reward	−1,098,613	[−1,180,137, −1,020,014]	$<10^{-3}$
H3	DQN − Static-threshold (week 6, early)	Profit	−1,032	[−3,407, +1,426]	0.414
H3	DQN − Static-threshold (week 25, mid-horizon)	Profit	+124,995	[+96,224, +153,126]	$<10^{-3}$